THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** cork and leaf	Cork and leaf are lighter than water.
2	**(d)** the boat's shape holds a lot of air	Large volume with air makes the boat's overall density small.
3	**(c)** food does not stick to the surface	Non-stick.
4	**(b)** salt and sugar	Both dissolve.
5	**(b)** wax melts quickly along the rod when one end is heated	Heat travels along it.
6	**(b)** wool is a poor conductor of heat	Traps body heat.
7	**(b)** metal	Metals have these properties.
8	**(d)** a thick wooden handle	Wood does not conduct heat.
9	**(b)** steel	Strong and waterproof.
10	**(a)** thin cotton cloth	Translucent cloth.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The boat has a large hollow shape with air (1); Water supports it as its average density is low (1).	1+1
12	No privacy (1); Frosted translucent glass is better (1).	1+1
13	Both disappear on stirring (1); Both are soluble (1).	1+1
14	Steel is stronger and harder than pure iron (1); Stainless steel does not rust easily (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnet removes iron (1); Dissolve sugar in water (1); Filter sand and evaporate water for sugar (1).	1+1+1
16	Rough mat cleans shoes (1); Smooth top is easy to clean (1); Window lets in light (1).	1+1+1
17	Thermocol is light (1); Combined object has low overall density (1); It floats (1).	1+1+1
	OR: Heat one end of each (1); Place wax pins at equal distance (1); The pin that falls first shows the best conductor (1).	1+1+1
18	Cut a piece of cork (1); Even small solid pieces float (1); Hence lighter material (1).	1+1+1
	OR: Pins conduct (1); Plastic insulates (1); Safe for users (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Heat water and cover (1); Measure temperature drop after 30 min (1); Same conditions for all (1); The least drop is best insulator (1); Cork or plastic likely (1).	1+1+1+1+1
	OR: Dissolve salt in water (1); Fold filter paper (1); Pour mixture (1); Sand stays on paper (1); Salt solution passes (1).	1+1+1+1+1
20	Heat a rod for conduction (1); Stir in water for solubility (1); Torch for transparency (1); Put in water for floating (1); Scratch for hardness (1).	1+1+1+1+1
	OR: Animal: wool, silk (1); Plant: cotton, wood (1); Mineral: iron, stone (1); Man-made: plastic (1); Natural sources give raw materials (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** wooden chips	Wood is non-magnetic.
2	**(c)** can be made into thin sheets	Metals are malleable.
3	**(a)** wood	Wood burns.
4	**(c)** cooking oil	Oils do not mix.
5	**(d)** breaks easily on hitting	Glass.
6	**(d)** glass	Glass breaks when hit.
7	**(a)** they form shadows	They block light.
8	**(c)** elastic and soft	Rubber rubs off marks.
9	**(b)** plastic	Plastic.
10	**(a)** spectacles	Clear view.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Heat is conducted along the rod (1); It reaches nearer points first (1).	1+1
12	Each part needs a different property (1); A pan has a metal body and plastic handle (1).	1+1
13	Wool traps air (1); Air is a poor conductor of heat (1).	1+1
14	It conducts electricity well (1); It is flexible and easy to draw (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Ceramic is a poor conductor (1); Does not burn fingers (1); Keeps tea hot (1).	1+1+1
16	Blocks light (1); Keeps heat out (1); Gives privacy (1).	1+1+1
17	Cloth for body (1); Plastic zip (1); Metal buckle (1).	1+1+1
	OR: Glass: transparent, breakable (1); Plastic: light, does not break (1); Plastic is cheaper (1).	1+1+1
18	Raincoat needs a waterproof material (1); Cooking pan needs a heat-conducting metal (1); Properties decide use (1).	1+1+1
	OR: Polyester absorbs less water (1); Water stays on the surface (1); Evaporates quickly (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pan: conducts heat (1); Tyre: elastic (1); Window: transparent (1); Bridge: strong (1); Raincoat: waterproof (1).	1+1+1+1+1
	OR: It breaks easily (1); It is heavy (1); Plastic is lighter (1); Metals are stronger (1); Choose by use (1).	1+1+1+1+1
20	Copper: wires (1); Aluminium: foil (1); Glass: windows (1); Rubber: tyres (1); Wood: furniture (1).	1+1+1+1+1
	OR: Conductors: copper (1); Uses: vessels, rods (1); Insulators: wood, plastic (1); Uses: handles, flasks (1); Choose based on need (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **Materials Around Us** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** flexible	Paper, wood.
2	**(c)** rubber	Rubber does not conduct electricity.
3	**(d)** bathroom windows	Provides privacy with some light.
4	**(c)** it gets soggy and weak	Paper absorbs water.
5	**(d)** strong and hard	Strong.
6	**(d)** it is lustrous and does not rust	Shiny and unreactive.
7	**(a)** a rubber tyre	Rubber is elastic and strong.
8	**(b)** absorbs sweat	Absorbs water.
9	**(c)** it is transparent but takes up space	Air is transparent.
10	**(b)** jute, nylon	Fibres are flexible.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Rubber gives good grip (1); It is flexible and durable (1).	1+1
12	Container: steel or plastic (1); Cloth: cotton; tyre: rubber (1).	1+1
13	Cotton is from plants (1); Nylon is made in factories (1).	1+1
14	Iron conducts heat quickly (1); Wood is a poor conductor (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Wood: easy to shape (1); Graphite: marks paper (1); Rubber: erases; metal ferrule holds it (1).	1+1+1
16	Wood is lighter than the same volume of water (1); Steel is heavier than the same volume of water (1); So the wooden ruler floats and the steel ruler sinks (1).	1+1+1
17	Natural: leather, wool, cotton (1); Man-made: plastic, nylon (1); Polythene is also man-made (1).	1+1+1
	OR: Material is the substance (1); Object has a shape and use (1); Same material can make different objects (1).	1+1+1
18	Add water and stir (1); Sugar dissolves (1); Chalk remains as sediment (1).	1+1+1
	OR: Pour water on it (1); Observe if it passes (1); Check if it soaks (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Metal base conducts heat (1); Plastic handle insulates (1); Each part has a job (1); Combined object is safe (1); Both properties are used (1).	1+1+1+1+1
	OR: Sorting by material (1); Metals recycled (1); Glass and paper recycled (1); Saves resources (1); Reduces waste (1).	1+1+1+1+1
20	Cork, wood, bottle float (1); Coin, stone, nail sink (1); Floating objects are lighter than water (1); Sinking are heavier (1); Shape can matter with hollow things (1).	1+1+1+1+1
	OR: Does not decay (1); Pollutes land and water (1); Jute bags (1); Paper and cloth bags (1); Reuse plastic (1).	1+1+1+1+1

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